

ARCHITECTURE BLUEPRINT

High-Level Design and Architecture

Azure-first, consent-led architecture for an explainable professional matching product

Style: Offline-capable modular monolith with managed Azure PaaS

Deployment: Single Malaysia-aligned region; dev and production environments

Quality attributes: Privacy, event resilience, explainability, maintainability and cost control

Review date: 27 August 2026

Architecture decision: Use one deployable API plus one asynchronous worker/job and one Azure SQL Database. Keep domain modules separated in code, but do not distribute them into microservices until real scaling or team boundaries appear.

1. Executive architecture summary

OLGA Connect is a bounded, consent-first professional matching system. A member becomes visible only inside a joined context, while presence is active and an intent has not expired. The system recommends at most seven explainable matches, opens messaging only after mutual acceptance, and captures the meeting outcome as the learning signal.

- React Native with Expo development builds for iOS and Android.
- Container Apps, Azure SQL, Entra External ID, Queue/Blob Storage, Notification Hubs and event-scale Web PubSub.
- Embeddings plus versioned deterministic rules; no trained ranking model or LLM explanation in v1.
- Precomputed context packages, local SQLite scoring and an idempotent mobile outbox.
- No coordinate history; store context membership and time-boxed presence only.

2. Scope resolution and assumptions

Blueprint tension	Resolved position
Malaysia vs India	Assume Kuala Lumpur / Malaysia for Release 1; confirm legal and data-region requirements before production.
Phone-only OTP	Use managed email OTP. If phone-only is fixed, run a separate managed-CIAM evaluation; never custom-build auth.
File sharing conflict	Defer file exchange and malware scanning; keep Blob abstractions for photos/exports/future files.
Geofence vs QR	QR/code is authoritative; geofence may suggest context but is not stored.
Offline learned model	Cache Azure-generated embeddings and run deterministic ranking on device.
Ledger in v1	Include it and snapshot intent at acceptance; it provides value at low network density.

3. Architecture principles

Principle	Design consequence
Consent is authoritative	Every match, disclosure, notification and conversation checks current consent/connection state.
Context is privacy boundary	Retrieval starts with context membership and LIVE presence; no cross-context discovery.
Expiry is enforced on read	Intent, presence and requests carry expires_at; delayed cleanup cannot revive them.
Explainability is a gate	Suppress any match lacking approved reason codes and a deterministic sentence.
Offline writes are normal	Mutations have idempotency keys, payload versions and outbox state.
Managed PaaS first	Avoid VMs, Kubernetes and self-hosted databases.
Preview isolation	Preview vector acceleration may be tested but cannot be a launch dependency.

4. Logical component architecture

OLGA Connect - Recommended Azure MVP topology

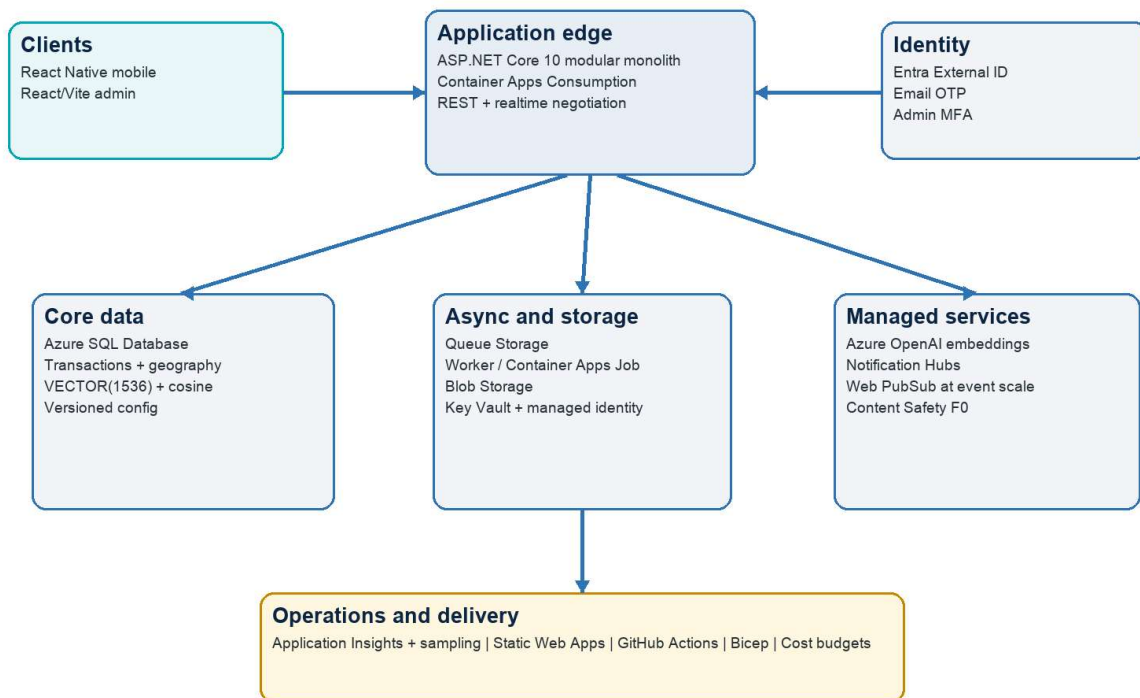


Figure 1. Recommended logical Azure topology for the MVP.

Component	Responsibilities	Scale boundary
Mobile app	Onboarding, intent, live view, requests, chat, ledger, cache/outbox.	Device and context-package size.
Admin portal	Verification, event setup, manual matches, moderation and live metrics.	Static hosting; API authorization.

Component	Responsibilities	Scale boundary
API modular monolith	REST, authz, domain rules, idempotency and realtime negotiation.	Stateless Container Apps replicas.
Worker/job	Embedding, match generation, expiry, push, outcomes and exports.	Queue depth and job concurrency.
Azure SQL	Transactions, geography, vectors, ranking config, audit and durable messages.	Serverless compute + relational indexes.
Azure OpenAI	Short normalized-intent embeddings only.	Token quota and deployment capacity.
Web PubSub	Event-scale chat and state notifications.	20 free connections; Standard for event.
Notification Hubs	APNs/FCM push for requests, acceptance and outcomes.	Free exploration; Basic >500 devices.

5. Domain modules

Module	Owns	Invariant
Identity/Member	External subject, member state, profile and verification	Suspended/deleted members cannot act or appear.
Context	Event/venue/org, membership and join lifecycle	Visibility never crosses contexts.
Intent/Presence	Active intent, snapshots and LIVE state	Both active and unexpired to match.
Matching	Eligibility, vectors, scores, reasons, feedback and manual source	Filters precede score; every card has a reason.
Connection	Request state, acceptance, block/report and snapshots	Conversation only after acceptance.
Conversation	Messages, delivery and moderation metadata	Authorize both participants on every object.
Meeting/Ledger	Outcome, usefulness, follow-up and derived ledger	Intent snapshot preserved at acceptance.
Administration	Roles, thresholds, config, manual match and audit	Actor, reason and timestamp for sensitive actions.

6. Core data design

Use one Azure SQL Database with schemas by module. Store timestamps in UTC, use rowversion for optimistic conflicts and durable unique constraints for idempotency.

Entity group	Key fields	Retention / privacy
Member/Profile	external_subject, status, public card fields, visibility	Minimize pre-acceptance fields; hard-delete or irreversibly anonymize by policy.
Context/Membership	type, window, join_method, member, role	Scoped membership; venue geometry is not user location history.
Intent	category, original/normalized text, expiry, status	Mandatory expiry; snapshot only for accepted relationships.
Embedding	intent, model, dimensions, hash, VECTOR(1536)	Delete with source unless lawfully retained; recompute on hash/model change.

Entity group	Key fields	Retention / privacy
Presence	member, context, went_live_at, expires_at	No latitude/longitude history; short retention.
Match	pair, component scores, reasons, rank version, source	Time-bounded identity-bearing record; de-identify evaluation facts.
Connection/Message	state, participants, body, delivery, moderation	Object-level authz and documented message retention.
Meeting/FollowUp	happened, usefulness, due_at, status	Private ledger data; export/deletion supported.
Consent/Safety/Audit	purpose, status, actor, subject, reason	Restricted, append-oriented evidence with approved retention.

7. Matching and NLP design

Explainable matching pipeline - v1

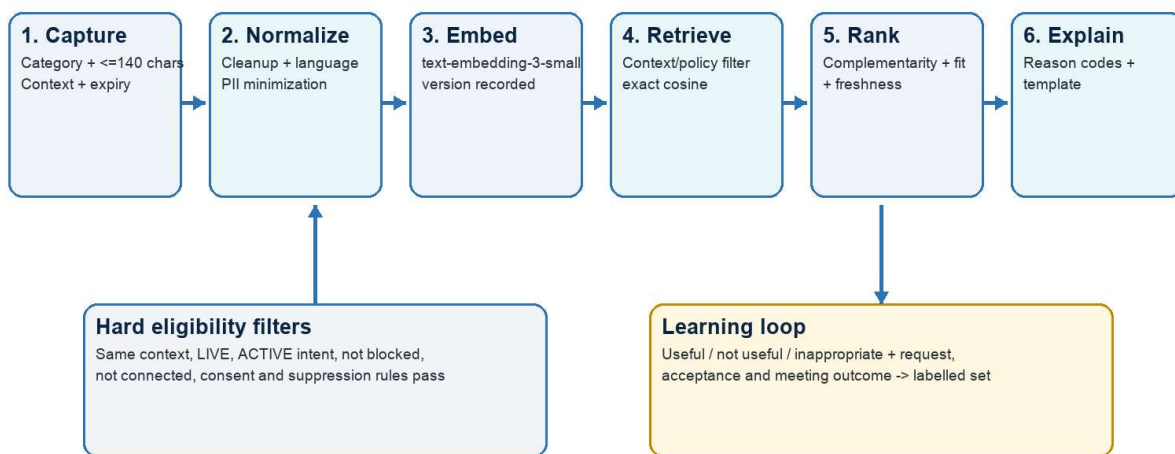


Figure 2. Observable v1 matching pipeline.

7.1 Eligibility

A pair is eligible only when both members are active, share one context, are LIVE, have ACTIVE unexpired intents, have not blocked each other, are not connected, are not the same member, and pass moderation/suppression policy. Recheck before card delivery and request creation.

7.2 Semantic representation

- Normalize Unicode/whitespace, add language tag and safe abbreviation map, minimize obvious PII, and preserve original text separately.
- Embed with text-embedding-3-small at 1536 dimensions; store model/deployment version and normalized-text hash.
- Use Azure SQL VECTOR(1536), float32, and exact cosine VECTOR_DISTANCE after context and policy filtering.
- Do not depend on approximate VECTOR_SEARCH preview for launch.

7.3 Reciprocal score

Illustrative v1 score: 0.40 category complementarity + 0.25 reciprocal semantic score + 0.15 industry/function adjacency + 0.10 seniority compatibility + 0.10 presence freshness. Store weights in a versioned table, not code.

Compute A-needs-to-B-offers and B-needs-to-A-offers. Use harmonic mean for mutual benefit; use a documented directional score for buyer/supplier categories. Cap shown matches at seven.

7.4 Explanation and evaluation

- Generate reason codes only from contributing signals and render approved templates.
- Record viewed, requested, accepted, met, usefulness, not-useful and inappropriate events with versions.
- Create at least 200 labelled pairs split by member or organization. Track Precision@5, Recall@20, reciprocal precision, acceptance and not-useful rate.
- Rollback by switching active ranking_config_id or embedding deployment after regression testing.

8. End-to-end flows

8.1 Onboarding and context join

1. Complete managed email OTP; API links member to immutable external subject.
2. Commit profile and active intent, then enqueue embedding work.
3. Worker normalizes, embeds and stores versioned vector.
4. Scan signed event QR; create membership without coordinate history.
5. Go Live creates a two-hour Presence and returns an expiring context package.

8.2 Match to conversation

6. API/device applies eligibility and scoring, then returns 3-7 explained cards.
7. Request uses an idempotency key; API rechecks consent, blocks, expiry and visibility.
8. Acceptance transaction creates connection, intent snapshots and conversation.
9. Message is authorized and durably stored before realtime fan-out.
10. Outcome prompt updates the private ledger and evaluation loop.

9. Offline-first event design

Highest technical risk: Crowded hotel Wi-Fi and captive portals are more likely to fail than Azure. After context join, the app must stay useful without a network.

Concern	Design
Context package	Signed, expiring set of public card fields, active intent snapshot, category metadata and precomputed vectors.
Local store	SQLite for cache, matches and outbox; secure storage for tokens/keys only; delete after retention window.
Offline ranking	Same versioned deterministic score with golden vectors matching the server.

Concern	Design
Outbox	idempotency_key, operation, payload version, timestamp, attempts and sync state.
Server replay	Unique user + idempotency key; state-machine validation; repeated call returns original result.
UX	Last-synced time and offline badge; never show empty truth or indefinite spinner.
Privacy	Current context and minimum fields only; wipe on logout/context expiry.
Rehearsal	Hundreds of seeds, throttled 3G, airplane mode, captive portal, duplicate retries and expiry.

10. Security, privacy and compliance

Area	Required design
Identity	Managed customer identity, short-lived tokens, admin MFA and separate admin roles.
Authorization	Server-side object checks on profiles, matches, requests, conversations and exports.
Location	Transient device geofence or QR; persist membership/presence, never coordinate history.
Consent	Purpose-specific records for presence, notifications and analytics with immediate revocation.
Protection	TLS, SQL/Storage encryption, managed identity, Key Vault and no CI/source secrets.
Abuse	Rate limits by subject/device/IP, block/report surfaces and audited staff review.
AI boundary	Only normalized intent reaches embeddings; never chat, precise location or unnecessary PII.
Data rights	Export, correction, withdrawal and hard deletion/anonymization with completion evidence.
Audit	Admin actions, consent, exports, suppression, manual matches and model/rank versions.
Legal	Qualified counsel confirms Malaysia PDPA, retention, cross-border processing and region.

11. Reliability, performance and scale

Attribute	Target / approach
Match feed	p95 <1.5s cached; pre-generate when intent/presence changes.
Write ack	p95 <500ms at event load; commit before fan-out.
Live view	<2s on mid-range Android over 3G; cache renders immediately.
Availability	Matching/chat may degrade; consent/authz never fail open.
API	Stateless replicas; tune connection pools and max scale from tests.
Worker	Queue concurrency, poison handling and failed-work queue.
Database	Normal indexes, Query Store, bounded vector scan; scale before event.
Realtime	Free for development only; Standard unit with buffer for event.
Observability	OpenTelemetry, structured logs, queue age, quality and consent dashboards.
Recovery	PITR/backups, restore drill and configuration rollback.

12. Deployment and delivery

Environment	Topology	Policy
Local	SQL Server 2025 Developer container, API/worker, emulator/device	Synthetic or anonymized data.
Development	Scale-to-zero API, SQL free offer, Static Web Apps Free, FO moderation	May pause; no SLA; safe for learning.
Production pilot	Separate resource group, paid-overflow SQL, managed identity, backups/alerts	Real data and approved retention.
Event mode	Warm API, raised SQL, Web PubSub Standard, Notification Hubs Basic	Load-tested runbook, freeze, rollback and scale-down.

- Provision with Bicep and Azure Developer CLI using environment parameter files.
- Deploy through GitHub Actions and OpenID Connect federation; no cloud client secret in GitHub.
- Use Container Apps revisions for health-checked rollout and rollback.
- Start without VNet/private endpoints unless the production security review requires them.

13. API contracts

Use versioned REST with OpenAPI. Every write accepts Idempotency-Key and returns a stable result. Use cursor pagination, rowversion/ETags, Problem Details and opaque IDs. WebSocket messages signal clients to refetch authoritative state.

API group	Representative operations
Member	GET /me; PATCH /me/profile; POST /me/export; DELETE /me
Contexts	POST /contexts/{id}/join; POST/DELETE /contexts/{id}/presence
Intents/Matches	POST/PATCH /intents; GET /matches; POST /matches/{id}/feedback
Connections	POST/PATCH /connection-requests; POST /blocks; POST /reports
Messaging	GET /conversations; GET/POST /conversations/{id}/messages
Ledger	POST /connections/{id}/outcome; GET /ledger; POST/PATCH /follow-ups
Admin	Events, verification, moderation, config, manual matches, dashboards and audit

14. Delivery gates

Stage	Deliverable	Gate
0 Decisions	Malaysia, email OTP, retention, moderation and QR priority	Owners approve before production provisioning.
1 NLP baseline	Anonymized data, 200 labels, matrix, notebook and metrics	Precision/error report and reasons.
2 Core	Identity, intent, context, match, request/accept and ledger	Authz, expiry and idempotency tests.

Stage	Deliverable	Gate
3 Offline	Context package, SQLite and outbox	Airplane mode and duplicate replay on iOS/Android.
4 Event ops	Push, realtime, staff console, seeds and outcomes	Load test and dress rehearsal.
5 Pilot	Small cohort and quality review	Severity-1 closed; config rollback rehearsed.

15. First architecture decision records

- ADR-001 Malaysia-first single-region deployment and residency assumptions.
- ADR-002 Email OTP and escalation path if phone-only remains mandatory.
- ADR-003 Azure SQL VECTOR(1536) exact search; no AI Search or preview ANN.
- ADR-004 QR-authoritative join and no coordinate-history storage.
- ADR-005 Offline package, local retention and idempotent outbox.
- ADR-006 Modular monolith boundaries and extraction criteria.
- ADR-007 Ranking config, reason catalog, evaluation ownership and rollback.
- ADR-008 File sharing deferred.

16. Top risks

Risk	Impact	Mitigation
Event network failure	No value at critical moment	Pre-onboarding, cache, offline score, outbox and rehearsal.
Low-quality intents	Weak matches	Guidance, examples, category + 140 chars and label review.
Phone-only mismatch	Delay/security risk	Approve email OTP or managed CIAM; no custom auth.
Location inference	Trust/compliance damage	QR-first, no coordinates, coarse presence and alert limits.
Preview vector dependency	Launch regression	GA vector/distance path; adapter for experiments.
Cost surprise	Budget erosion	Sampling, caps, budgets, scale-to-zero and event runbook.
Cold start	Slow first interaction	Warm one replica during event and pre-test.
Sparse network	Empty matches/churn	Seed profiles, manual matches and ledger value.

Reference sources

Reviewed 27 August 2026. Azure capabilities, quotas and prices vary by region and agreement; confirm in the Azure pricing calculator before purchase.

[Azure SQL Database free offer FAQ](#)

[Azure SQL vector functions](#)

[Azure SQL vector data type](#)

[Azure OpenAI embeddings tutorial](#)

[Azure Container Apps billing](#)

[Azure Container Apps scaling](#)

[Microsoft Entra External ID authentication methods](#)

[Microsoft Entra External ID pricing concepts](#)

[Azure Web PubSub pricing and capacity](#)

[Azure Notification Hubs FAQ](#)

[Azure Static Web Apps plans](#)

[Azure AI Content Safety overview](#)

[Azure Monitor daily cap](#)

[.NET support policy](#)

[React Native environment guidance](#)